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SkillRise: Agentic Reinforcement Learning for Cross-Task Skill Evolution

A Single Policy Alternates Between Solving Tasks and Curating a Growing Skill Document—Rewarded for Downstream Impact, Not Document Plausibility

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Standard agentic RL treats each task as an independent episode, discarding transferable workflows after every interaction. **SkillRise** trains a single policy to simultaneously solve tasks and maintain an evolving natural-language skill document. A key innovation is *decoupled credit assignment*: skill curation is rewarded based on its downstream impact on future tasks, not on document quality alone. On ALFWorld, WebShop, and ScienceWorld, SkillRise achieves the best Pass@1 among all compared methods.

AT A GLANCE

Problem: Agentic RL discards generalizable patterns after each task, preventing efficient transfer to future related tasks.

Why it matters: Real-world agents face streams of related tasks; the ability to curate and reuse skills is essential for continual improvement.

Method: Single policy that solves tasks and curates a skill document; decoupled credit assignment rewards curation by downstream task improvement.

Result: Best Pass@1 on ALFWorld (78.9%), WebShop (66.3%), ScienceWorld (50.2%).

Evidence: 11–12 pp. gain over standard RLVR; shared solve-curate policy outperforms separate-policy baselines.

The Big Picture

Agentic reinforcement learning for LLMs (RLVR) has demonstrated that language agents can learn to navigate web interfaces, plan household tasks, and execute scientific workflows purely from outcome-based rewards. But these

agents are trained and tested on episodic tasks: each episode is independent, and the knowledge gained in one episode is only preserved insofar as it is absorbed into the model’s parameters through gradient updates.

This means every new task starts from scratch. A household agent that learned to find items in a drawer on task 1 has no mechanism to record that workflow and apply it directly on task 2. SkillRise proposes a simple but effective alternative: let the agent write its own workflow documentation, and reward the documentation by how much it helps future tasks.

Why Existing Approaches Fall Short

Prior work on skill-augmented agentic RL (Voyager, SKILL1, SkillRL, SkillFlow) decomposes skill evolution into a pipeline:

1. *Extract*: A separate skill extraction module reads completed trajectories and identifies transferable patterns.
2. *Store*: Extracted skills are written to a retrieval database.
3. *Retrieve*: At test time, a retriever fetches relevant skills as context.

This pipeline has two weaknesses. First, it requires separate optimization objectives for each component: extraction, retrieval, and execution are trained separately with potentially conflicting objectives. Second, the skill document is static after extraction and cannot be revised in light of future task failures—a pattern extracted from an easy instance may mislead on a harder variant.

SkillRise collapses all three stages into a single reinforcement learning objective applied to a single policy.

Core Mechanism

SkillRise operates over a sequence of N tasks $T_1, T_2, \dots, T_N$ organized from easy to hard. For each task T_t , the agent is given the current skill document S_{t-1} and takes two sequential actions:

Solve: Generate a trajectory $a_{1:H}$ to complete T_t given context (T_t, S_{t-1}) .

Curate: Given $(T_t, a_{1:H}, S_{t-1})$, generate an updated skill document S_t that adds, revises, or deletes workflow entries in light of what was learned on T_t .

Both actions are produced by the same policy π_θ ; no separate extraction or retrieval module exists.

Technical Formulation

Let $r_{\text{solve}}(T_t, a_{1:H})$ be the task completion reward and $r_{\text{curate}}(S_t, T_{t+1:t+k})$ be the discounted impact of the updated skill document on the next k tasks:

$$r_{\text{curate}}(S_t, T_{t+1:t+k}) = \frac{1}{k} \sum_{j=1}^k \gamma^j \cdot \Delta r_{\text{solve}}(T_{t+j}, S_t \text{ vs. } S_{t-1})$$

The SkillRise objective is:

$$\mathcal{J}(\theta) = \mathbb{E} \left[\sum_{t=1}^N r_{\text{solve}}(T_t, a_{1:H}^{(t)}) + \gamma \cdot r_{\text{curate}}(S_t, T_{t+1:t+k}) \right]$$

This is the critical distinction from prior work: curation is rewarded by downstream impact (Δr_{solve}), not by surface-level quality. A skill is only reinforced when it demonstrably helps solve future tasks.

Learning or Inference Procedure

Training uses Group Relative Policy Optimization (GRPO): multiple curation outputs are sampled for each post-task state, and the policy is updated based on relative advantage between outputs that have better vs. worse downstream impact.

The training curriculum organizes tasks by difficulty within each domain: easier instances first. This ensures that skill documents accumulate useful content early in training before harder tasks demand more nuanced workflows. Skill documents are constrained to 2,000 tokens to fit within the context window alongside task instructions.

What the Guarantee Says

Decoupled credit assignment provides a monotonicity property: if the curation policy is improved, the downstream solve reward is non-decreasing in expectation. This follows from the definition of r_{curate} : if S_t improves performance on future tasks, the curation reward is positive, incentivizing the policy to generate more useful updates.

For a fixed backbone, the SkillRise objective upper-bounds the cumulative task completion rate over the N -task stream.

Experimental Findings

ALFWorld (household task completion, Pass@1):

Method	Pass@1
Base LLM	42.3%
Standard RLVR	67.8%
SKILL1	71.4%
SkillRise	78.9%

WebShop (e-commerce navigation, Task Success):

Method	Success
Standard RLVR	56.2%
SkillFlow	61.4%
SkillRise	66.3%

ScienceWorld (scientific experiment tasks, Pass@1):

Method	Pass@1
Standard RLVR	38.1%
SKILLC	43.7%
SkillRise	50.2%

Gains over standard RLVR are 11.1 pp (ALFWorld), 10.1 pp (WebShop), and 12.1 pp (ScienceWorld).

Ablations and Interpretation

Decoupled vs. immediate curation reward: Replacing r_{curate} with an immediate quality score (document fluency or coverage) reduces ALFWorld Pass@1 by 6.6 pp, confirming that downstream impact is the essential signal.

Shared vs. separate policy: Training a separate curation policy (independent weights) reduces ALFWorld Pass@1 by 4.1 pp, suggesting productive weight-sharing: solving improves curation, and curation improves solving through shared world model knowledge.

Curriculum ordering: Random task ordering reduces ALFWorld Pass@1 by 6.3 pp, confirming that the easy-to-hard curriculum is essential for early skill accumulation before difficult tasks arrive.

Document length: Allowing up to 4,000 tokens (vs. 2,000) gains 1.2 pp on ScienceWorld, suggesting the constraint is occasionally binding for complex scientific workflows.

Reference

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